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# Predictive Modelling with Linear Regression: ShowTime

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# 1. Introduction

In today's highly competitive digital streaming landscape, content performance is evaluated from the very first day of release. First-day viewership serves as a crucial performance indicator, reflecting immediate audience interest, marketing effectiveness, and platform engagement levels. Strong opening performance not only enhances visibility within the platform's recommendation systems but also contributes to sustained user engagement, word-of-mouth promotion, and long-term content success.

For ShowTime, optimizing first-day content performance is strategically important. With increasing competition from other streaming platforms and alternative entertainment sources, understanding the factors that drive audience behavior has become essential. Various internal and external elements may influence first-day viewership. These include promotional efforts such as trailer views and advertising impressions, overall platform traffic levels, timing of release (day of the week and season), content genre preferences, and competing events such as major sports tournaments that may divert audience attention.

While marketing investments and scheduling decisions are often based on experience and intuition, data-driven insights provide a more reliable foundation for strategic planning. By analyzing historical performance data, ShowTime can identify which factors significantly influence first-day content views and which variables have limited impact. This enables the organization to allocate resources more effectively, optimize release strategies, and enhance overall return on investment.

In this study, a predictive modeling approach using linear regression has been applied to evaluate the relationship between first-day content views and multiple explanatory variables. The objective is twofold: first, to develop a statistically reliable model capable of predicting first-day performance; and second, to generate actionable business insights that can guide future content launch decisions. Through systematic exploratory analysis, model building, assumption testing, and performance evaluation, this report aims to provide strategic recommendations that support improved decision-making and stronger content outcomes.

## 2. Objective of the Study

The primary objective of this study is to identify and quantify the key factors that influence first-day content views on the ShowTime platform. Understanding these drivers is essential for improving content launch strategies, optimizing marketing investments, and enhancing overall platform performance.

Specifically, the study aims to:

- Analyze the relationship between first-day content views and various independent variables, including trailer views, ad impressions, platform visitors, release timing, seasonality, genre, and major sports events.
- Develop a predictive model using linear regression to estimate first-day content performance based on historical data.
- Evaluate the statistical significance and impact of each predictor to determine which factors meaningfully influence opening-day results.
- Assess the model's performance using appropriate evaluation metrics to ensure reliability and generalizability.
- Derive actionable insights that can guide strategic decisions related to marketing planning, release scheduling, and content promotion.

Through this analysis, the study seeks to move beyond intuition-based decision-making and provide a data-driven framework that supports more effective content launch strategies at ShowTime.

# 3. Methodology

To achieve the objectives of this study, a structured analytical approach was followed, beginning with data exploration and ending with model evaluation.

First, an exploratory data analysis (EDA) was conducted to understand the distribution of variables, identify patterns, and detect potential issues such as missing values, outliers, and skewness. Visualizations such as histograms, boxplots, and scatter plots were used to examine relationships between variables and the target variable (first-day content views).

Next, data preprocessing steps were performed to prepare the dataset for modeling. Outliers in continuous variables were identified using the Interquartile Range (IQR) method and treated using capping to reduce the impact of extreme values while preserving all observations. Categorical variables such as genre, day of week, and season were converted into numerical format using one-hot encoding to make them suitable for regression analysis.

The dataset was then split into training and testing sets using an 80:20 ratio. The training set was used to build the linear regression model, while the testing set was used to evaluate its predictive performance on unseen data.

A multiple linear regression model was developed using the training data to examine the relationship between first-day content views and the selected independent variables. Model assumptions including linearity, independence of errors, normality of residuals, homoscedasticity, and multicollinearity were tested to ensure the validity of the results.

Finally, the model's performance was evaluated using metrics such as R-squared, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics helped assess both explanatory power and prediction accuracy.

This structured methodology ensures that the findings are both statistically sound and practically relevant for business decision-making.

# 4. Data Description

The dataset consists of 1000 observations, where each observation represents a single content release on the ShowTime platform. The variables included in the dataset capture marketing efforts, platform activity, release timing, and external conditions that may influence first-day content performance.

Below is a description of the key variables used in the analysis:

## Target Variable

- views_content: represents the number of views a piece of content received on its first day of release. This is the dependent variable that the model aims to predict.

## Numerical Independent Variables

- visitors: Total number of users visiting the platform around the time of release. This reflects overall platform traffic.
- ad_impressions: Number of advertisement impressions generated for the content. This represents the scale of advertising exposure.
- views_trailer: Number of views received by the promotional trailer before release. This indicates pre-release audience engagement.
- major_sports_event: A binary variable (0 = No, 1 = Yes) indicating whether a major sports event was occurring at the time of release.

## Categorical Independent Variables

- genre: Category of the content (e.g., Drama, Comedy, Action, etc.).
- dayofweek: The day on which the content was released.
- season: The season during which the content was launched (Spring, Summer, Fall, Winter).

The dataset includes both continuous and categorical variables, allowing for a comprehensive analysis of how marketing factors, timing decisions, and external events influence first-day content performance.

# 5. Exploratory Data Analysis

## 5.1 Initial Data Inspection

### 5.1.1 Viewing the First Five Records

```
   visitors  ad_impressions  major_sports_event     genre  dayofweek  season  \
0      1.67         1113.81                   0    Horror  Wednesday  Spring
1      1.46         1498.41                   1  Thriller     Friday    Fall
2      1.47         1079.19                   1  Thriller  Wednesday    Fall
3      1.85         1342.77                   1    Sci-Fi     Friday    Fall
4      1.46         1498.41                   0    Sci-Fi     Sunday  Winter

   views_trailer  views_content
0          56.70           0.51
1          52.69           0.32
2          48.74           0.39
3          49.81           0.44
4          55.83           0.46
```

The first five records show a mix of numerical variables (visitors, ad_impressions, views_trailer, views_content, major_sports_event) and categorical variables (genre, dayofweek, season).

The values vary across observations, indicating that the dataset captures different marketing intensities, release timings, and performance levels. The structure appears clean, with no obvious formatting inconsistencies.

### 5.1.2 Viewing the Last Five Records

```
     visitors  ad_impressions  major_sports_event    genre  dayofweek  season  \
995      1.58         1311.96                   0  Romance     Friday    Fall
996      1.34         1329.48                   0   Action     Friday  Summer
997      1.62         1359.80                   1   Sci-Fi  Wednesday    Fall
998      2.06         1698.35                   0  Romance     Monday  Summer
999      1.36         1140.23                   0   Comedy   Saturday  Summer

     views_trailer  views_content
995          48.58           0.36
996          72.42           0.56
997         150.44           0.66
998          48.72           0.47
999          52.94           0.49
```

The last five records confirm that the dataset maintains consistency throughout. Variable formats remain uniform, and there are no structural irregularities between the beginning and end of the dataset.

The variability in performance metrics across rows suggest that the dataset contains sufficient diversity to support meaningful modeling and analysis.

## 5.2 Checking Data Dimensions

```
(1000, 8)
```

The dataset contains 1000 rows and 8 columns, indicating 1000 content release observations.

Out of these, 7 variables act as predictors (visitors, ad_impressions, major_sports_event, genre, dayofweek, season, views_trailer) and 1 variable (views_content) serves as the target variable.

The dataset size is sufficient to build a reliable regression model, as it provides a reasonable number of observations relative to the number of predictors.

## 5.3 Viewing Column Names

```
Index(['visitors', 'ad_impressions', 'major_sports_event', 'genre',
       'dayofweek', 'season', 'views_trailer', 'views_content'],
      dtype='object')
```

The dataset contains the following variables:

- visitors: Number of platform visitors.
- ad_impressions: Number of advertising impressions.
- major_sports_event: Indicator of whether a major sports event occurred (0/1)
- genre: Content genre category
- dayofweek: Day on which the content was released
- season: Season of release
- views_trailer: Number of trailer views prior to release
- views_content: First-day content views (**Target Variable**)

Among these:

- Numerical Variables: visitors, ad_impressions, views_trailer, views_content
- Categorical Variables: genre, dayofweek, season
- Binary Variable: major_sports_event

The structure clearly separates marketing metrics, timing variables, and performance outcomes, making it suitable for regression modeling.

## 5.4 Data Background and Content

The dataset consists of 1000 observations representing individual content releases on the platform. Each observation captures marketing activity, platform conditions, release characteristics, and the resulting first-day performance.

The objective of the dataset is to understand which factors influence first-day content views and to build a predictive model based on these drivers.

| # | Column | Non-NULL Count | Dtype |
|---|---|---|---|
| 0 | visitors | 1000 non-NULL | float64 |
| 1 | ad_impressions | 1000 non-NULL | float64 |
| 2 | major_sports_event | 1000 non-NULL | int64 |
| 3 | genre | 1000 non-NULL | object |
| 4 | dayofweek | 1000 non-NULL | object |
| 5 | season | 1000 non-NULL | object |
| 6 | views_trailer | 1000 non-NULL | float64 |
| 7 | views_content | 1000 non-NULL | float64 |

The dataset contains 8 variables, categorized as follows:

- Performance Variable (Target)
  - views_content: Number of first-day content views.

- Marketing & Engagement Variables
  - ad_impressions: Number of advertising impressions.
  - views_trailer: Number of trailer views before release.

- Platform Activity Variable
  - visitors: Number of platform visitors.

- External Influence Variable
  - major_sports_event: Binary indicator (1 = sports event occurred, 0 = no event)

- Release Characteristics
  - genre: Genre of the content.
  - dayofweek: Day of release.
  - season: Season during which content was released.

All 1000 observations are complete with no missing values. The dataset includes a mix of numerical and categorical variables, making it suitable for regression modeling after appropriate preprocessing.

## 5.5 Checking for Missing Values

```
visitors              0
ad_impressions        0
major_sports_event    0
genre                 0
dayofweek             0
season                0
views_trailer         0
views_content         0
dtype: int64
```

- All variables show zero missing values, indicating that the dataset is complete across all 1000 observations.
- There are no null values in:
  - Marketing metrics (ad_impressions, views_trailer)
  - Platform activity (visitors)
  - Release characteristics (genre, dayofweek, season)
  - External factor (major_sports_event)
  - Target variable (views_content)
- Since no missing data is present, no imputation or data cleaning for null values is required.

## 5.6 Checking for Statistical Summary

- The descriptive statistics provide insight into the distribution and spread of the numerical variables.
  - Visitors have a mean of 1.70, ranging from 1.25 to 2.34, indicating moderate variability in platform traffic across releases.
  - Ad impressions show wider dispersion, with values ranging from approximately 1,011 to 2,424 and a standard deviation of 289.53, suggesting varying levels of marketing intensity.
  - Trailer views range from 30 to nearly 200, with a mean of 66.91. The relatively higher maximum compared to the median (53.96) suggests the presence of high-performing outliers.
  - Major sports event has a mean of 0.40, indicating that around 40% of releases occurred during a major sports event.

- For the target variable:
  - First-day content views (views_content) have a mean of 0.473 and a median of 0.45.
  - The range extends from 0.22 to 0.89, showing noticeable variation in opening performance.
  - The relatively moderate standard deviation (0.106) indicates performance varies but remains within a manageable range.

- Overall, the statistics suggest sufficient variability in predictors to support regression modeling while maintaining stability in the target variable.

## 5.7 Checking for and treating (if needed) data irregularities

### 5.7.1 Checking for duplicate records

- The result shows 0 duplicate rows.
- Each record represents a unique content release.
- There is no repetition that could inflate counts or distort model results.
- No action is required for duplicate removal.

### 5.7.2 Checking for logical inconsistencies

- The variable **major_sports_event** contains only valid values (0 and 1), so the binary encoding is correct.
- There are no cases where **first-day content views** exceed trailer views, which maintains logical consistency between promotional engagement and actual performance.
- The **genre** column includes valid categories such as Horror, Thriller, Sci-Fi, Drama, Action, Comedy, Romance, and Others.
- The **dayofweek** variable includes all seven days with no unexpected labels.
- The **season** variable contains only Spring, Fall, Winter, and Summer, no irregular or misspelled entries.

### 5.7.3 Checking for invalid or unrealistic

- No negative values were found in any of the numerical variables.
- All values fall within realistic and logically acceptable ranges.
- Visitors vary within a stable range, indicating consistent platform traffic levels.
- Ad impressions show wider variation, which reflects differences in marketing intensity across releases.
- Trailer views have some higher maximum values, suggesting a few highly promoted or highly anticipated titles.
- First-day content views remain within a reasonable range, with moderate spread and no extreme anomalies.

There are no unrealistic or suspicious numerical entries in the dataset. The data appears valid and suitable for further analysis.

### 5.7.4 Checking for Consistency of Categorical Data

- No extra spaces were found in the genre column; category labels are clean.
- Case formatting is consistent across genres, dayofweek, and season.
- All expected categories are present:
    - Genres include Horror, Thriller, Sci-Fi, Drama, Action, Comedy, Romance, and Others.
    - All seven days of the week are represented.
    - Seasons include Spring, Fall, Winter, and Summer.
- Category counts show a reasonably distributed dataset:
    - The “Others” genre has the highest count, but no genre appears suspiciously low or duplicated.
    - Days of the week and seasons are also fairly balanced.

Overall, the categorical variables are clean, consistent, and do not require correction before encoding.

## 5.8 Univariate Analysis

### 5.8.1 Identifying Numerical & Categorical Variables

```
Numerical Variables:
Index(['visitors', 'ad_impressions', 'major_sports_event', 'views_trailer',
       'views_content'],
      dtype='object')

Categorical Variables:
Index(['genre', 'dayofweek', 'season'], dtype='object')
```

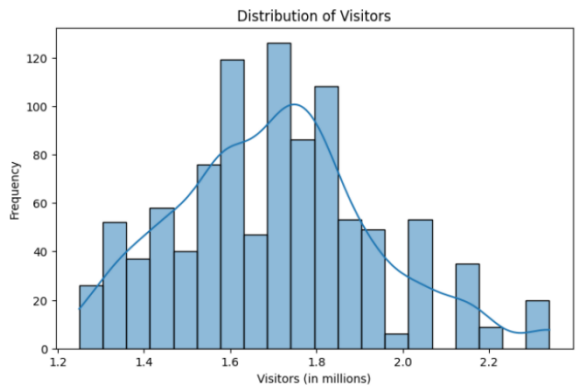
- The dataset clearly separates into numerical and categorical variables.
- Numerical variables include **visitors**, **ad_impressions**, **major_sports_event**, **views_trailer** and **views_content**.
- Categorical variables include **genre**, **dayofweek**, and **season.**

This makes things straightforward. The numerical variables can directly be used in regression modeling, while the categorical variables will require encoding.

Overall, the data structure looks organized and easy to work with, no confusion between variable types.

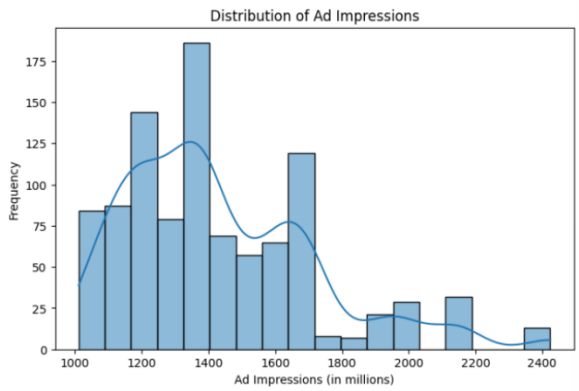
### 5.8.2 Exploring all Numerical Variables

1. Visitors



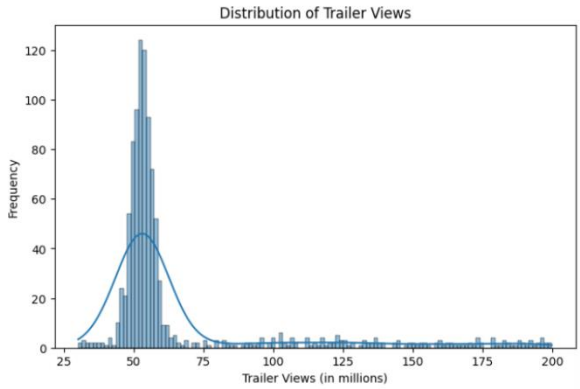

- The distribution of visitors looks fairly close to normal, with most values concentrated between roughly 1.5 and 1.9 million.
- The skewness value (~0.37) indicates a slight right skew, meaning there are a few higher traffic days pulling the distribution slightly to the right.
- There are no extreme spikes or unusual gaps in the distribution.

2. Ad Impressions




- The distribution of ad impressions is clearly right-skewed.
- The skewness value (~1.03) confirms a strong positive skew, meaning a small number of releases received significantly higher advertising exposure than the rest.
- Most values are concentrated between roughly 1100 and 1700 impressions, with a long tail extending toward 2400.

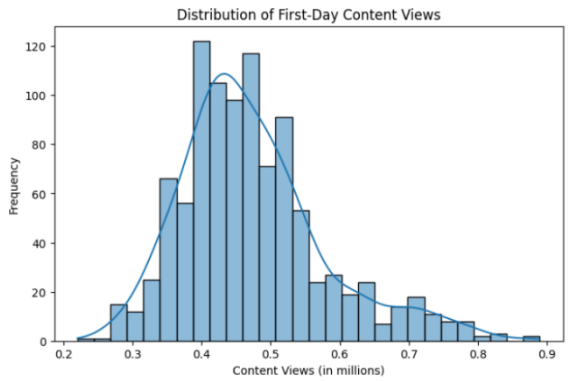
3. Trailer Views




- The distribution of trailer views is highly right-skewed.
- The skewness value (~2.37) confirms a strong positive skew, meaning a small number of titles received exceptionally high trailer engagement.
- Most content is clustered tightly around 45-60 trailer views, while a few releases extend far beyond 100 and even close to 200.

4. Content Views

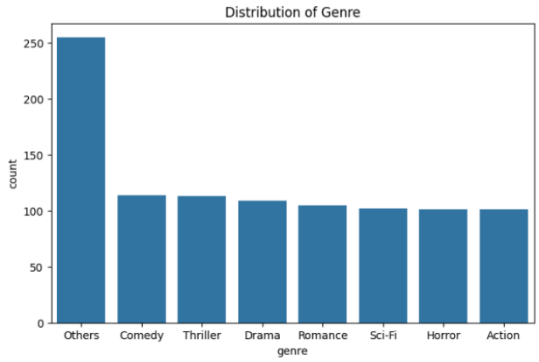
- The distribution of first-day content views is moderately right-skewed.
- The skewness (~0.94) indicates a noticeable positive skew, meaning a few titles perform significantly better than the average.
- Most releases fall within the 0.35 to 0.55 range, showing fairly consistent performance for the majority of content.
- A smaller group of titles reaches much higher view counts (above 0.70), creating the long right tail.




## 5.8.3 Exploring Categorical Variables

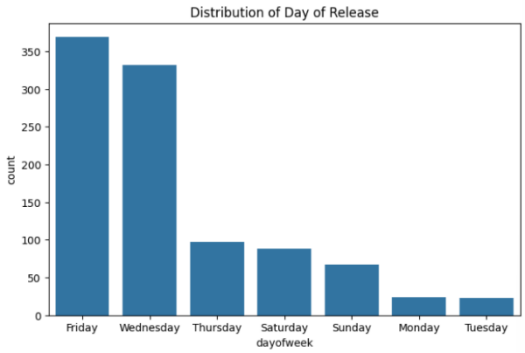
1. Genre

- The "Others" category has the highest share, accounting for about 25.5% of the dataset.
- The remaining genres (Comedy, Thriller, Drama, Romance, Sci-Fi, Horror, Action) are fairly evenly distributed, each contributing roughly 10-11%.
- No single genre (apart from "Others") dominates the dataset.



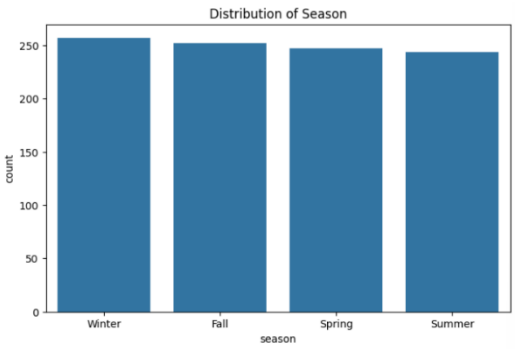

2. Day of Week

- A large share of releases happen on Friday (36.9%) and Wednesday (33.2%). Together, they account for more than two-thirds of all releases.
- Very few releases occur on Monday (2.4%) and Tuesday (2.3%), making them the least preferred days.
- Thursday, Saturday, and Sunday have moderate but noticeably lower proportions compared to Friday and Wednesday.



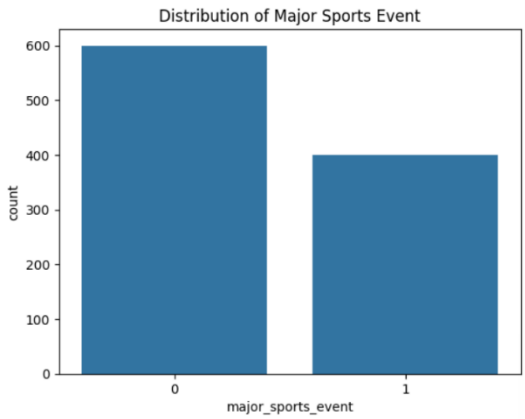

3. Season

- The distribution across seasons is almost perfectly balanced.
- Winter (25.7%), Fall (25.2%), Spring (24.7%), and Summer (24.4%) each contribute roughly one-fourth of the total releases.
- No single season dominates the dataset.




4. Major Sports Event

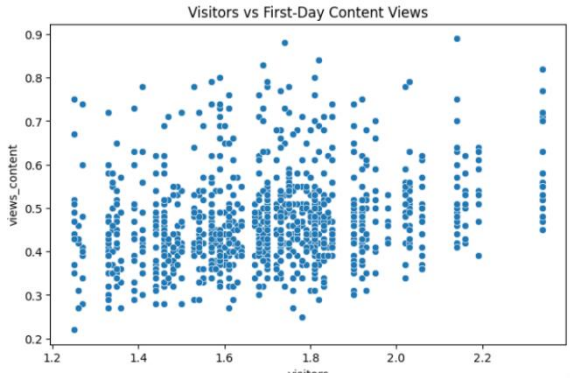
- A majority of content releases (60%) happened when there was no major sports event, while 40% coincided with a major sports event.
- This shows a slight imbalance, but both scenarios are still well represented in the data.
- The presence of a sizeable 40% during sports events suggests that the platform does not completely avoid releasing content on such days.



## 5.9 Bivariate Analysis

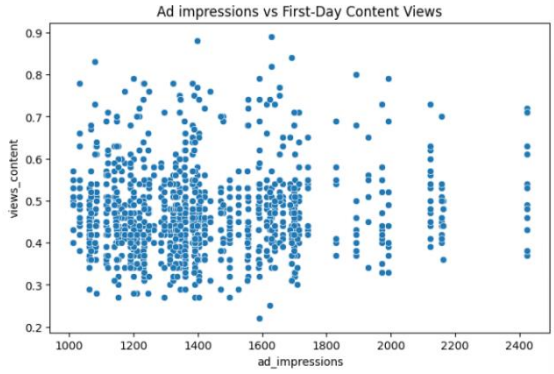
### 5.9.1 Numerical Variables vs Target Variable (views_content)

1. Visitors vs views_content



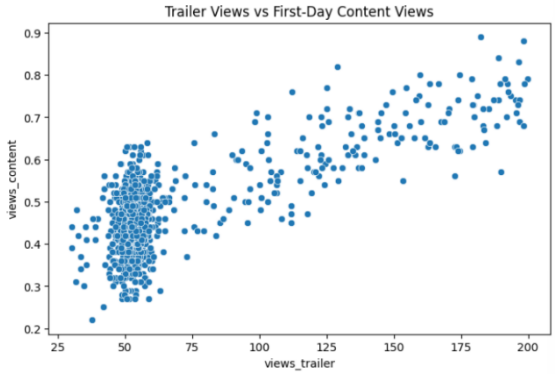

- The scatter plot shows a positive relationship between visitors and first-day content views. As platform traffic increases, content views also tend to increase.
- However, the relationship is not very strong, the correlation coefficient is 0.26, which indicates a weak to moderate positive association.
- The points are fairly scattered, meaning visitors alone do not fully explain variations in first-day performance.

2. Ad Impressions vs First-Day Content Views




- The scatter plot does not show any clear pattern between ad impressions and first-day content views.
- The correlation coefficient is 0.05, which is extremely weak and almost negligible.
- Even at higher levels of ad impressions, content views vary widely, indicating inconsistent impact.

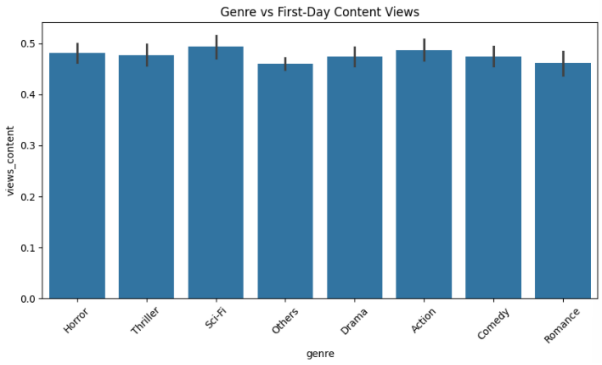
3. Trailer Views vs First-Day Content Views




- The scatter plot clearly shows a strong upward trend, as trailer views increase, first-day content views also increase.
- The correlation coefficient is 0.75, which indicates a strong positive relationship.
- Unlike ad impressions, this relationship looks structured and consistent, not random.

### 5.9.2. Categorical Variables vs Target Variable (views_content)

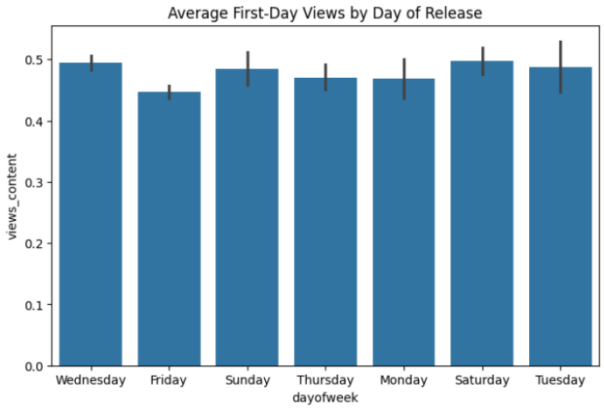
1. Genre vs First-Day Content Views




- The average first-day content views across genres are fairly close to each other, mostly ranging between 0.46 and 0.50 million.
- No single genre dramatically outperforms the others.
- Sci-Fi and Action appear slightly higher on average, while Romance and Others are slightly lower, but the differences are small.

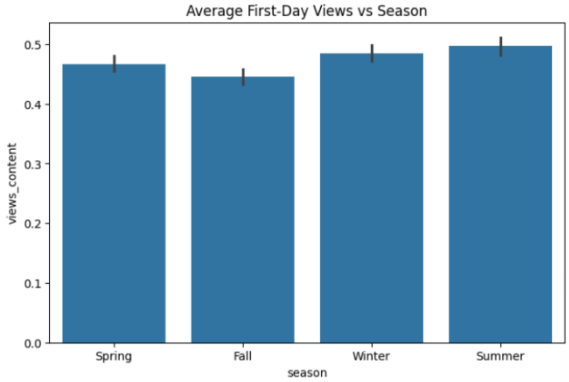
2. First-Day Content Views vs Day of Release

- First-day views are slightly higher for releases on Wednesday and Saturday, both touching close to 0.50 million on average.
- Friday actually appears a bit lower compared to mid-week and weekend releases.
- The differences are present but not extremely large, most days fall within a narrow range.



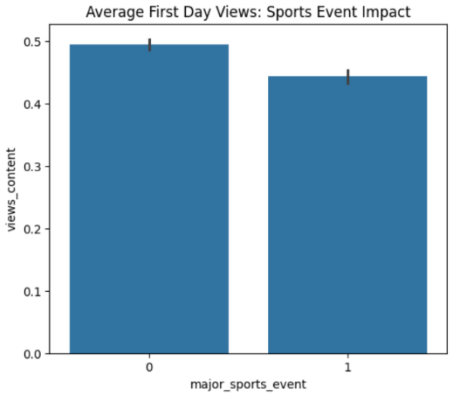

3. First-Day Content Views vs Season

- First-day views are highest in Summer and Winter, both close to 0.50 million on average.
- Fall shows slightly lower performance compared to the other seasons.
- The differences are not extreme, but there is a visible seasonal pattern.




4. Average First-Day views: Sports Event Impact

- When there is no major sports event (0), average first-day views are clearly higher, close to 0.50 million.
- When a major sports event is happening (1), first-day views drop noticeably to around 0.44-0.45 million.
- The gap is quite visible, not just a tiny fluctuation.




## 5.10 Answers to the Key Questions provided

### 5.10.1. What does the distribution of content views look like?

The distribution of first-day content views looks slightly right-skewed. Most pf the values are concentrated between 0.35 and 0.55 million views, forming a fairly smooth, bell-shaped pattern. However, there is a noticeable tail stretching toward higher values (up to around 0.9 million), which pulls the distribution slightly to the right. The skewness value (around 0.94) confirms that it is not perfectly normal, but its not extremely skewed either.

Thus, we can conclude:

- Most content performs in a moderate range on day one.
- A smaller number of releases perform exceptionally well, creating that right-side tail.

So overall, performance is fairly stable with a few standout hits driving higher numbers.

### 5.10.2. What does the distribution of genres look like?

The distribution of genres is mostly balanced, but not perfectly equal.

- The “Others” category stands out the most, making up about 25.5% of the dataset which is clearly the largest share.
- The remaining genres - Comedy, Thriller, Drama, Romance, Sci-Fi, Horror, and Action which are fairly evenly distributed, each contributing roughly 10-11%.

So apart from “Others” being slightly dominant, the dataset doesn’t heavily favor any one specific genre.

Thus, the platform seems to have a diverse mix of content types, which is good because it reduces genre bias in the analysis.

### 5.10.3. How does viewership vary with the day of release?

Viewership does vary across the days of the week, but the differences are moderate rather than drastic.

From the analysis:

- Wednesday and Saturday show the highest average first-day views.
- Friday performs slightly lower than expected.
- Monday and Tuesday tend to have comparatively lower average viewership.
- The remaining days fall within a similar performance range.

Overall, while the day of release does have some impact on first-day performance, the variation is not extremely large. This suggests that release timing plays a role, but it is not the primary driver of viewership compared to factors like trailer engagement or platform traffic.

### 5.10.4. How does viewership vary with the season of release?

Viewership does show some variation across seasons, though the differences are not very large.

From the analysis:

- Summer and Winter have slightly higher average first-day views compared to the other seasons.
- Spring performs moderately.
- Fall shows comparatively lower average viewership.

The variation is noticeable but not extreme, all seasons fall within a fairly close range of performance.

This suggests that season does have some influence on first-day viewership, possibly due to changes in audience availability or viewing habits, but it is not the strongest factor driving performance.

### 5.10.5. What is the correlation between trailer views and content views?

The correlation between trailer views and first-day content views is 0.75.

This indicates a strong positive relationship. When trailer views increase, first-day content views tend to increase as well, and quite consistently.

A correlation of 0.75 is relatively high, which means trailer engagement is closely linked to launch-day performance. It suggests that trailers are not just promotional tools, they are strong indicators of audience interest and potential success.

## 5.11. Key Insights

- ➢ Trailer views have a strong positive correlation (0.75) with first-day content views, making trailer engagement the strongest predictor of launch performance.
- ➢ Platform visitors show a positive but moderate relationship with content views, meaning higher traffic helps but is not the only driver.

- Ad impressions have almost no meaningful correlation with first-day views, suggesting that advertising volume alone does not guarantee success.
- Major sports events negatively impact first-day performance, indicating that audience attention shifts during major sporting events.
- Release day shows some variation in performance, with Wednesday and Saturday performing slightly better on average.
- Seasonal differences exist, with Summer and Winter showing marginally higher average viewership.
- Genre does not appear to significantly differentiate first-day performance; average views are fairly similar across categories.
- The distribution of content views is moderately right-skewed, meaning most titles perform within a stable range, while a few outperform significantly.
- The dataset is clean, balanced, and free from missing values or logical inconsistencies, making it suitable for reliable modeling.

# 6. Data Preprocessing

## 6.1 Checking for and treating (if needed) Outliers

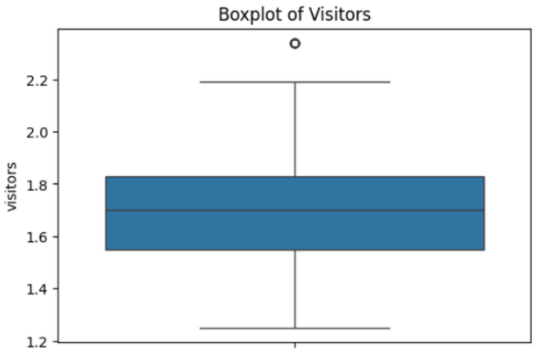
### 6.1.1. Identifying Numerical Columns

```
Numerical Columns:
Index(['visitors', 'ad_impressions', 'major_sports_event', 'views_trailer',
       'views_content'],
      dtype='object')
```

- The dataset contains five numerical variables.
- These are: visitors, ad_impressions, major_sports_event, views_trailer, and views_content.
- visitors, ad_impressions, views_trailer, and views_content are continuous numerical variables.
- major_sports_event is a binary numerical variable (0 = No event, 1 = Event).
- The numerical features include both engagement metrics and an event indicator, making them suitable for further statistical analysis and modeling.

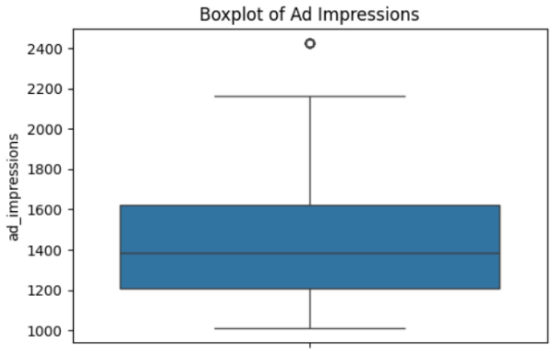
### 6.1.2 Visual Inspection of Outliers using Boxplot

**1. Checking for Outliers in the "visitors" column**



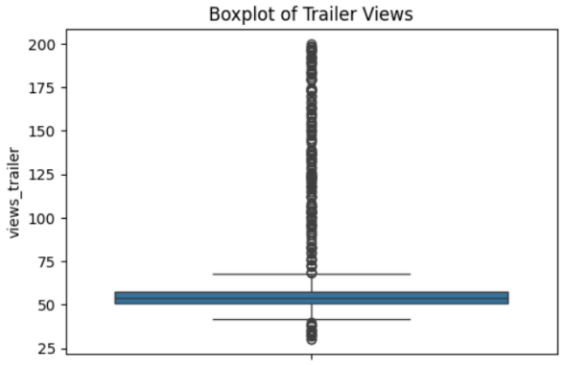

- The median number of visitors is around 1.7 million, indicating that most releases see traffic around this level.
- The middle 50% of values (IQR) lies roughly between 1.55 and 1.83 million, showing moderate variation.
- There is at least one upper outlier above 2.3 million visitors.
- The spread is slightly wider on the upper side, suggesting mild right skewness.

**2. Checking for Outliers in the "ad impressions" column**




- The median ad impressions are around 1,380-1,400 million, indicating the typical campaign exposure level.
- The middle 50% of values lie roughly between 1,200 and 1,620 million, showing noticeable spread.
- There is a clear upper outlier above 2,400 million impressions.
- The upper whisker is longer than the lower side, suggesting a right-skewed distribution.

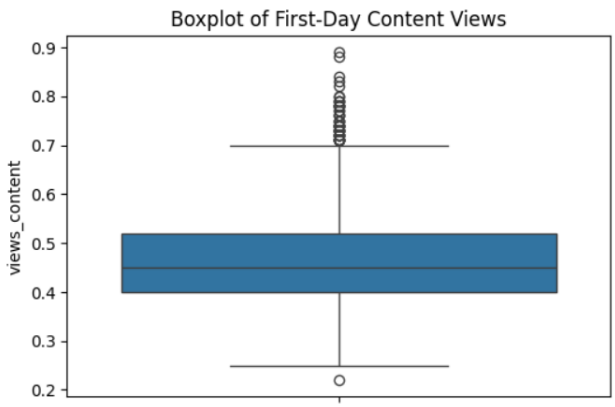
**3. Checking for Outliers in the "trailer views" column**




- The median trailer views are around 53-55 million, meaning most content clusters around this range.
- The IQR is very tight, roughly between 50 and 58 million, showing that the majority of trailers perform similarly.
- However, there are many upper outliers, extending all the way up to nearly 200 million views.
- There are also a few lower outliers below 40 million.

**4. Checking for Outliers in the "content views" column**

- The median first-day content views are around 0.45 million, which represents the typical launch-day performance.
- The middle 50% of values lie roughly between 0.40 and 0.52 million, showing moderate consistency in performance.
- There are several upper outliers, reaching up to around 0.9 million, indicating a few standout hits.
- There is also a small lower outlier near 0.22 million.




### 6.1.3. Detecting Outliers through IQR Method

```
visitors: 20 outliers detected
ad_impressions: 13 outliers detected
views_trailer: 189 outliers detected
views_content: 47 outliers detected
```

- **views_trailer** has the highest number of outliers (189), indicating a heavy upper tail and the presence of many viral-level trailers.
- **views_content** has 47 outliers, showing that while most content performs within a stable range, a few releases significantly outperform others.
- **visitors** has 20 outliers, suggesting occasional spikes in platform traffic.
- **ad_impressions** has 13 outliers, meaning most campaigns operate within a fairly consistent range.

### 6.1.4 Treating Outliers

1. Treating outliers in "visitors" column:-

- After applying IQR-based clipping to the *visitors* column, the number of outliers reduced to 0.
- This means all extreme values have been capped within the acceptable lower and upper bounds.
- The treatment preserves the overall distribution while preventing unusually high or low traffic spikes from distorting the model.

2. Treating outliers in "ad_impressions" column:-

- After applying IQR-based clipping to the *ad_impressions* column, the number of outliers reduced to 0.
- All extreme impression values have been capped within the calculated lower and upper limits.
- This ensures that unusually high advertising campaigns do not disproportionately influence the regression model.

3. Treating outliers in "views_trailer" column:-

- After applying IQR-based clipping to the *views_trailer* column, the number of others reduced to 0.
- The previously large number of extreme trailer view values has now been capped within acceptable bounds.
- This significantly reduces the impact of viral-level trailers that could otherwise distort the regression model.

4. Treating outliers in "views_content" column:-

- After applying IQR-based clipping to the *views_content* column, the number of outliers reduced to **0**.
- All extreme first-day performance values have been capped within acceptable statistical limits.
- This ensures that unusually high-performing or low-performing releases do not disproportionately influence the regression results.

## 6.2 Feature Engineering

### 6.2.1. Identifying Categorical Variables

```
Categorical Variables:
Index(['genre', 'dayofweek', 'season'], dtype='object')
```

- The dataset contains three categorical variables: *genre, dayofweek,* and *season*.
- These variables represent qualitative attributes related to content classification and release timing.
- Since they are stored as object type, they require encoding before being used in regression modeling.
- These features help capture behavioral patterns linked to content type and timing effects.

### 6.2.2. Applying One-Hot Encoding

```
   visitors  ad_impressions  major_sports_event  views_trailer  views_content  \
0      1.67         1113.81                   0          56.70           0.51
1      1.46         1498.41                   1          52.69           0.32
2      1.47         1079.19                   1          48.74           0.39
3      1.85         1342.77                   1          49.81           0.44
4      1.46         1498.41                   0          55.83           0.46

   genre_Comedy  genre_Drama  genre_Horror  genre_Others  genre_Romance  ...  \
0         False        False          True         False          False  ...
1         False        False         False         False          False  ...
2         False        False         False         False          False  ...
3         False        False         False         False          False  ...
4         False        False         False         False          False  ...

   genre_Thriller  dayofweek_Monday  dayofweek_Saturday  dayofweek_Sunday  \
0           False             False               False             False
1            True             False               False             False
2            True             False               False             False
3           False             False               False             False
4           False             False               False              True

   dayofweek_Thursday  dayofweek_Tuesday  dayofweek_Wednesday  season_Spring  \
0               False              False                 True           True
1               False              False                False          False
2               False              False                 True          False
3               False              False                False          False
4               False              False                False          False

   season_Summer  season_Winter
0          False          False
1          False          False
2          False          False
3          False          False
4          False           True
```

- The categorical variables (*genre, dayofweek, season*) have been successfully converted into numerical dummy variables using one-hot encoding.
- Each category is now represented as a separate binary column (True/False).
- The original categorical columns have been replaced with their encoded versions.
- This transformation makes the dataset fully numerical and suitable for regression modeling.

### 6.2.3. Verifying the Transformed Dataset

```
Columns after encoding:
Index(['visitors', 'ad_impressions', 'major_sports_event', 'views_trailer',
       'views_content', 'genre_Comedy', 'genre_Drama', 'genre_Horror',
       'genre_Others', 'genre_Romance', 'genre_Sci-Fi', 'genre_Thriller',
       'dayofweek_Monday', 'dayofweek_Saturday', 'dayofweek_Sunday',
       'dayofweek_Thursday', 'dayofweek_Tuesday', 'dayofweek_Wednesday',
       'season_Spring', 'season_Summer', 'season_Winter'],
      dtype='object')
```

- After encoding, the dataset now contains only numerical columns.

- The original categorical variables have been replaced with their respective dummy variables (for genre, dayofweek, and season)
- One category from each group is dropped (baseline category) to avoid multicollinearity.
- The dataset is now fully prepared for regression modeling, with all predictors in numerical form.

## 6.3 Data Preparation for Modeling

### 6.3.1. Defining Independent (X) and Dependent (Y) Variables

```
Shape of X: (1000, 20)
Shape of Y: (1000,)
```

- The feature matrix contains 1000 rows and 20 predictor variables.
- The target variable Y contains 1000 observations, matching the number of rows in X.
- This confirms there is no mismatch between predictors and the dependent variable.
- The dataset is properly structured and ready for train-test splitting and model building.

### 6.3.2. Train-Test Split

```
Training set size: (800, 20)
Testing Set Size: (200, 20)
```

- The dataset has been split into 800 training observations and 200 testing observations.
- This follows an 80:20 split, which is a standard and balanced approach.
- Both training and testing sets contain 20 predictor variables, ensuring consistency.
- The training set will be used to build the model, while the testing set will evaluate its performance on unseen data.

### 6.3.3. Adding Constant for Linear Regression

```
     const  visitors  ad_impressions  major_sports_event  views_trailer  \
29     1.0     1.25         1710.10                   1       54.49000
535    1.0     1.60         1336.16                   0       55.66000
695    1.0     1.72         1390.62                   0       67.96625
557    1.0     1.92         1294.78                   0       55.84000
836    1.0     1.67         1010.87                   1       53.09000

     genre_Comedy  genre_Drama  genre_Horror  genre_Others  genre_Romance  \
29          False        False         False          True          False
535         False        False         False          True          False
695         False         True         False         False          False
557         False        False         False         False          False
836          True        False         False         False          False

     ...  genre_Thriller  dayofweek_Monday  dayofweek_Saturday  \
29   ...           False             False                True
535  ...           False             False               False
695  ...           False             False               False
557  ...            True             False                True
836  ...           False             False               False

     dayofweek_Sunday  dayofweek_Thursday  dayofweek_Tuesday  \
29              False               False              False
535             False               False              False
695             False               False              False
557             False               False              False
836             False               False              False

     dayofweek_Wednesday  season_Spring  season_Summer  season_Winter
29                 False           True          False          False
535                False          False          False           True
695                False          False          False          False
557                False           True          False          False
836                 True           True          False          False
```

- A constant column (*const*) has been successfully added to the training and testing datasets.
- This represents the intercept term required for linear regression using statsmodels.

- The feature structure remains intact, with all 20 predictors plus the constant term.
- The dataset is now correctly formatted for fitting the OLS regression model.

# 7. Model Building- Linear Regression

## 7.1 Building the Linear Regression Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.622
Model:                            OLS   Adj. R-squared:                  0.612
Method:                 Least Squares   F-statistic:                     64.05
Date:                Sat, 28 Feb 2026   Prob (F-statistic):          2.85e-149
Time:                        21:06:56   Log-Likelihood:                 1109.8
No. Observations:                 800   AIC:                            -2178.
Df Residuals:                     779   BIC:                            -2079.
Df Model:                          20
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     -0.2852      0.028    -10.010      0.000      -0.341      -0.229
visitors                   0.1280      0.010     13.249      0.000       0.109       0.147
ad_impressions          3.758e-06   7.72e-06      0.487      0.626   -1.14e-05    1.89e-05
major_sports_event        -0.0564      0.005    -12.472      0.000      -0.065      -0.048
views_trailer              0.0091      0.000     28.859      0.000       0.009       0.010
genre_Comedy              -0.0028      0.010     -0.293      0.770      -0.022       0.016
genre_Drama                0.0048      0.010      0.504      0.614      -0.014       0.024
genre_Horror              -0.0009      0.010     -0.094      0.925      -0.020       0.018
genre_Others              -0.0020      0.008     -0.236      0.813      -0.018       0.014
genre_Romance             -0.0094      0.010     -0.979      0.328      -0.028       0.009
genre_Sci-Fi               0.0151      0.010      1.517      0.130      -0.004       0.035
genre_Thriller             0.0063      0.010      0.650      0.516      -0.013       0.025
dayofweek_Monday           0.0355      0.015      2.371      0.018       0.006       0.065
dayofweek_Saturday         0.0453      0.008      5.367      0.000       0.029       0.062
dayofweek_Sunday           0.0384      0.009      4.238      0.000       0.021       0.056
dayofweek_Thursday         0.0213      0.008      2.796      0.005       0.006       0.036
dayofweek_Tuesday          0.0376      0.015      2.428      0.015       0.007       0.068
dayofweek_Wednesday        0.0444      0.005      8.495      0.000       0.034       0.055
season_Spring              0.0227      0.006      3.666      0.000       0.011       0.035
season_Summer              0.0403      0.006      6.368      0.000       0.028       0.053
season_Winter              0.0279      0.006      4.504      0.000       0.016       0.040
==============================================================================
Omnibus:                        8.940   Durbin-Watson:                   2.066
Prob(Omnibus):                  0.011   Jarque-Bera (JB):                9.119
Skew:                           0.251   Prob(JB):                       0.0105
Kurtosis:                       2.855   Cond. No.                     2.02e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 2.02e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

- The model explains about 62% of the variation in first-day content views ($R^2 = 0.622$, Adjusted $R^2 = 0.612$). This is a reasonably strong fit for real-world data.
- The overall model is statistically significant (Prob F-statistic ≈ 0.000), meaning the predictors collectively explain the target variable well.

Significant Predictors ($p < 0.05$):

- **views_trailer** has a strong positive impact and the highest t-value, clearly the most influential predictor.
- **visitors** positively affects first-day views.
- **major_sports_event** has a significant negative impact, confirming that sports events reduce viewership.
- Several **day-of-week variables** (Monday, Tuesday, Wednesday, Thursday, Saturday, Sunday) are significant.
- All **season variables (Spring, Summer, Winter)** are significant and positively associated with views.

Not Significant:

- **ad_impressions** is not statistically significant.
- Most **genre variables** are not significant predictors.

Diagnostics:

- Durbin-Watson ≈ 2.06 suggests no serious autocorrelation.
- The condition number is relatively high, which may indicate mild multicollinearity (likely due to dummy variables).

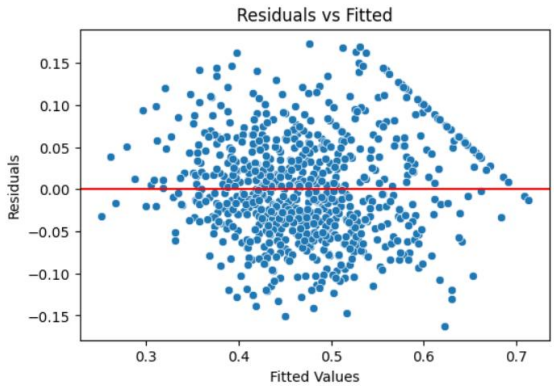
## 7.2. Creating a Dataframe of coefficients

| | Variable | Coefficient | P-value |
|---|---|---|---|
| visitors | visitors | 0.128048 | 2.797074e-36 |
| dayofweek_Saturday | dayofweek_Saturday | 0.045309 | 1.058855e-07 |
| dayofweek_Wednesday | dayofweek_Wednesday | 0.044417 | 9.961731e-17 |
| season_Summer | season_Summer | 0.040321 | 3.265043e-10 |
| dayofweek_Sunday | dayofweek_Sunday | 0.038447 | 2.528213e-05 |
| dayofweek_Tuesday | dayofweek_Tuesday | 0.037574 | 1.541416e-02 |
| dayofweek_Monday | dayofweek_Monday | 0.035546 | 1.800530e-02 |
| season_Winter | season_Winter | 0.027858 | 7.675032e-06 |
| season_Spring | season_Spring | 0.022693 | 2.627319e-04 |
| dayofweek_Thursday | dayofweek_Thursday | 0.021326 | 5.305617e-03 |
| genre_Sci-Fi | genre_Sci-Fi | 0.015115 | 1.297146e-01 |
| views_trailer | views_trailer | 0.009128 | 3.999554e-125 |
| genre_Thriller | genre_Thriller | 0.006257 | 5.157378e-01 |
| genre_Drama | genre_Drama | 0.004825 | 6.141775e-01 |
| ad_impressions | ad_impressions | 0.000004 | 6.263417e-01 |
| genre_Horror | genre_Horror | -0.000904 | 9.248593e-01 |
| genre_Others | genre_Others | -0.001960 | 8.133342e-01 |
| genre_Comedy | genre_Comedy | -0.002816 | 7.697527e-01 |
| genre_Romance | genre_Romance | -0.009373 | 3.278619e-01 |
| major_sports_event | major_sports_event | -0.056424 | 1.115396e-32 |
| const | const | -0.285167 | 2.842248e-22 |

- **Visitors** has the highest positive coefficient, meaning higher platform traffic directly increases first-day content views.
- **Day of week** variables (especially Saturday and Wednesday) show strong positive effects, suggesting timing does influence performance.)
- **Season variables**, particularly Summer and Winter, also positively impact first-day views.
- **views_trailer** is highly significant with a strong positive impact, reinforcing that trailer engagement drives launch success.
- **major_sports_event** has a negative coefficient and is highly significant, confirming that sports events reduce content viewership.
- **ad_impressions** has an almost negligible coefficient and is not statistically significant, indicating that ad volume alone does not meaningfully affect first-day performance.
- Most **genre variables** are not statistically significant, suggesting genre alone does not strongly determine launch-day success.

# 8. Testing the assumptions of Linear Regression Model

## 8.1 Testing Linearity




- The residuals are scattered randomly around the zero line, which is a good sign.
- There is no clear pattern (like a curve or funnel shape), suggesting the linearity assumption is reasonably satisfied.
- The spread of residuals looks fairly consistent across fitted values, indicating approximate homoscedasticity.
- A few larger residuals exist, but nothing extreme or systematically structured.

Overall, the residual plot suggests that the linear regression model fits the data reasonably well without major violations of assumptions.

## 8.2 Testing Independence of Errors (Durbin-Watson Test)

```
Durbin-Watson Statistic: 2.065683662184044
```

- The Durbin-Watson statistic is 2.07, which is very close to 2.
- A value around 2 indicates no significant autocorrelation in the residuals.
- This means the independence of errors assumption is satisfied.

## 8.3 Testing Homoscedasticity (Constant Variance)

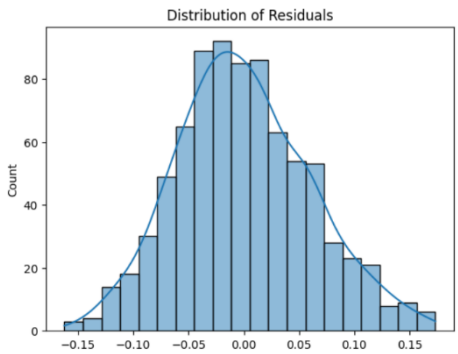
- The **LM test p-value (~ 0.000002)** is extremely small (less than 0.05).
- The **F-test p-value (~ 0.000001)** is also highly significant.
- Since both p-values are far below 0.05, we reject the null hypothesis of constant variance.

This means there is **evidence of heteroscedasticity**, the variance of residuals is not perfectly constant.

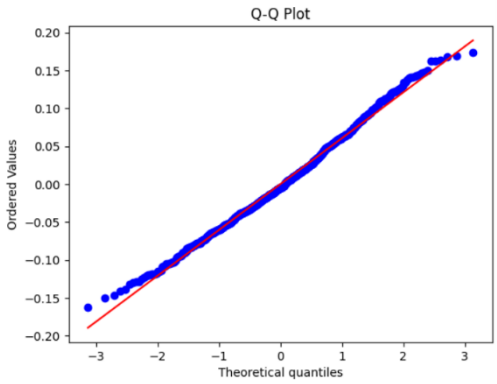
Thus, the spread of errors changes across fitted values. While the model still works, the presence of heteroscedasticity suggests that using robust standard errors would make the inference more reliable.

## 8.4 Testing Normality of Residuals

- The residuals form a roughly **bell-shaped distribution**, centered close to zero.
- The shape looks fairly symmetric, with only slight right skew.
- There are no extreme heavy tails or unusual spikes.



This suggests that the **normality assumption of residuals is reasonably satisfied**. While it may not be perfectly normal, it is close enough for linear regression to remain reliable.




- The points lie very close to the red reference line across most of the distribution.
- There are only slight deviations at the extreme ends (tails).
- This indicates that the residuals are approximately normally distributed.

Overall, the Q-Q plot supports the normality assumption of the linear regression model, with only minor tail deviations that are not severe.

## 8.5 Testing Multicollinearity

| | Feature | VIF |
|---|---|---|
| 0 | const | 173.064218 |
| 8 | genre_Others | 2.744608 |
| 7 | genre_Horror | 1.957056 |
| 11 | genre_Thriller | 1.952752 |
| 9 | genre_Romance | 1.951451 |
| 6 | genre_Drama | 1.949123 |
| 5 | genre_Comedy | 1.893001 |
| 10 | genre_Sci-Fi | 1.841998 |
| 19 | season_Summer | 1.570450 |
| 20 | season_Winter | 1.559595 |
| 18 | season_Spring | 1.536915 |
| 17 | dayofweek_Wednesday | 1.291416 |
| 15 | dayofweek_Thursday | 1.166108 |
| 13 | dayofweek_Saturday | 1.134732 |
| 14 | dayofweek_Sunday | 1.123843 |
| 16 | dayofweek_Tuesday | 1.062378 |
| 12 | dayofweek_Monday | 1.054613 |
| 3 | major_sports_event | 1.051825 |
| 4 | views_trailer | 1.023142 |
| 1 | visitors | 1.022671 |
| 2 | ad_impressions | 1.016695 |

- The VIF values for almost all predictors are close to 1, which indicates very low multicollinearity.
- All variables (except the constant) have VIF values well below 5, meaning there is no serious multicollinearity concern.
- The high VIF for the constant (173) is not an issue, which is common and can be ignored.

Overall, the predictors are not highly correlated with each other, and the model does not suffer from multicollinearity problems.

## 8.6 Findings from the Testings the Assumption of Linear Regression

### 8.6.1. Linearity

- The residuals vs fitted plot shows no clear curve or systematic pattern.
- Residuals are randomly scattered around zero.
- This suggests the linearity assumption is reasonably satisfied.

### 8.6.2 Independence of Errors

- The Durbin–Watson statistic is **2.07**, which is very close to 2.
- This indicates no significant autocorrelation.
- The independence assumption is satisfied.

### 8.6.3 Homoscedasticity (Constant Variance)

- The Breusch-Pagan test returned very small p-values (< 0.05).
- This means we reject the null hypothesis of constant variance.
- There is no evidence of heteroscedasticity.
- This model is still usable, but using robust standard errors would improve reliability.

### 8.6.4 Normality of Residuals

- The histogram shows an approximately bell-shaped distribution.
- The Q-Q plot points lie close to the straight line.
- Residuals are approximately normally distributed.

### 8.6.5 Multicollinearity

- VIF values for all predictors (except the constant) are close to **1**.
- No serious multicollinearity issues exist.
- The predictors are stable and not strongly correlated with each other.

# 9. Model Performance Evaluation

```
Model Performance on Test Data:
R-squared: 0.6044223215782114
MAE: 0.04895690433272268
MSE: 0.003991749629349367
RMSE: 0.06318029462854195
```

- The R-squared is ~0.60, which means the model explains about 60% of the variation in first-day content views on unseen data. That's actually pretty solid for real-world behavioral data.
- The MAE is ~0.049, meaning on average, predictions are off by about 0.05 units (in millions). So roughly ~50,000 views error on average. That's quite reasonable.
- The RMSE is ~0.063, which is slightly higher than MAE — this tells us there are a few larger errors, but nothing extreme.
- Since the test $R^2$ (~0.60) is close to the training $R^2$ (~0.62), the model is not overfitting. That's a good sign — it generalizes well.

## 9.1 Evaluating the model on difference performance Metrics

### 9.1.1 R-Squared ($R^2 \approx 0.604$)

- The model explains about 60% of the variation in first-day content views.
- That's a strong result for content performance data, which is naturally noisy.
- Since training $R^2$ (~0.62) and test $R^2$ (~0.60) are close, the model generalizes well.
- No signs of overfitting.

### 9.1.2 Mean Absolute Error (MAE ≈ 0.049)

- On average, predictions are off by about 0.049 million views.
- That's roughly 49,000 views error per prediction.
- MAE is easy to interpret and not overly sensitive to large errors.

### 9.1.3 Mean Squared Error (MSE ≈ 0.00399)

- Measures average squared prediction error.
- Penalizes larger mistakes more heavily.
- Value is low, which suggests no extreme prediction failures.

### 9.1.4 Root Mean Squared Error (RMSE ≈ 0.063)

- RMSE brings error back to the original scale.
- The model's prediction error is about 0.063 million views (~63,000 views).
- Slightly higher than MAE, meaning a few moderate outliers exist.

# 10. Actionable Insights & Recommendations

## 10.1 Significance of Predictors

Based on the regression results:

- Trailer views is one of the strongest and most significant predictors of first-day content views. This indicates that pre-release engagement directly influences opening performance.
- Visitors has the highest positive coefficient and is highly significant. This shows that overall platform strongly impacts content and consumption.
- Major Sports Events have a significant negative impact on first-day views. This suggests that audience shifts during major sporting events.
- Day of Release significantly affects performance. Several days (Saturday, Wednesday, Sunday, etc.) show higher expected first-day views compared to the baseline.
- Seasonality matters, with Summer showing the strongest positive effect on performance.
- Ad Impressions and most Genre variables are not statistically significant, indicating that ad volume and content category alone do not strongly determine first-day performance after controlling for other factors.

## 10.2 Key Business Takeaways

### 10.2.1 Focus on Trailer Engagement:

Since trailer views strongly drive first-day performance, the company should:

- Invest in high-impact trailer campaigns.
- Optimize trailer visibility on the platform.
- Promote trailers aggressively before release. Improving trailer engagement can directly increase opening-day success.

### 10.2.2 Optimize Release Timing:

- Avoid launching major content during large sports events.
- Prefer Summer releases when possible.
- Consider strategic day selection for release to maximise visibility.

Timing plays a measurable role in performance.

### 10.2.3 Increase Platform Traffic:

Since overall visitors strongly influence views:

- Run broader traffic-driving campaigns before major releases.
- Use homepage banner and recommendations to boost traffic. More traffic → higher content consumption.

### 10.2.4 Reconsider Ad Spending Strategy:

Ad impressions are not a significant driver in the current model. This suggests:

- Quality of engagement may matter more than ad volume.
- Marketing efforts should focus on meaningful engagement rather than just increasing ad exposure.

# 11. Conclusion

Overall, the model provides a clear and practical understanding of what drives first-day content views. The analysis shows that engagement-related factors, especially trailer views, play the most powerful role in determining performance. Content that generates strong trailer engagement is far more likely to translate that interest into actual first-day viewership. Platform traffic also significantly contributes, meaning that timing releases during high visitor activity periods can meaningfully improve outcomes.

The study also highlights the importance of release timing. Certain days of the week and particular seasons positively influence viewership, while releasing during major sports events negatively impacts performance. This reinforces the idea that audience attention is limited and highly sensitive to external events and behavioral patterns. On the other hand, variables such as genre and raw ad impressions do not independently drive first-day success once other factors are controlled. This suggest that simply **Major Sports Events** have a significant negative impact on first-day views. This suggests that audience attention shifts during major sporting events.

From a statistical perspective, the model is stable and reliable. It explains around 60% of the variation in first-day views, performs consistently on test data, and satisfies most regression assumptions, with only mild heteroscedasticity observed. Multicollinearity is not a concern, and residual diagnostics indicate that the model structure is appropriate.

In conclusion, first-day content success is primarily driven by audience engagement and strategic timing rather than surface-level characteristics. The findings provide meaningful direction for decision-making, helping stakeholders focus on improving trailer impact, aligning releases with favourable periods, and avoiding competitive attention clashes. While no predictive model is perfect, this one offers strong explanatory power and practical value for forecasting and strategic planning.